Class I

Transportation Engg

1) Highway

2) Railway

3) Dock & harbour

4) Airport & tunnel

Highway Engg →

Highway engg—

1) Introduction

2) Geometric design.

3) Sight distances

4) traffic engineering

5) Material engineering & material survey

6) Road characteristic.

7) Intersection or signal design.

8) Highway capacity & miscallaneous.

Highway Engg

Introduction

Highway developement & plan

Roads kis order me develop hui?

1) Roman Roads
2) Tresguard
3) Metcalfe
4) Telford road method
5) Macadam road method
6) CBR method.

(sequence is imp)

Road developement committees in India.

kya kya committe hai road ke developement ke liye india me

1) Jaykar committee. — 1927
2) IRF (Central road fund) — 1929
3) Indian road congress IRC (1934)
4) Motor vehicle act MVA (1939)
5) 1st 20 year road plan.
6) CRRI [Central road research institute] 1950
7) NHA National highway act
8) 2nd 20 year road plan
9) Highway research board.

(1.4)

- 20 yr road plan
  - 1st
    - year
    - Name
    - target
    - Names of road given
    - Pattern?
  - 2nd
    - Year
    - Name
    - kya introduce kiye?
    - target
  - 3rd
    - Year
    - Name
    - Roads classification kaise kiya
    - target

20 year road plan

1st 20 yr road plan.

- 1943 - 1963
- Nagpur road plan.
- length 16 km / 100 km² area.
- NH, SH, MDR, ODR, VR
  (major, district road, state highway, other, village road)
- Star & grid pattern / Radial & grid pattern

2nd 20 yr road plan

- 1961 - 1981
- Bombay road plane
- Expressway introduced + NH SH MDR OR VR
- 32 km / 100 km² area.

3rd 20 yr road plan.

- 1981 - 2001
- Lucknow road plan
- primary 1° < EW, NW
  secondary 2° < SH, MDR
  tertiary 3° ODR, VR

Requirement of ideal road kya hai.

Highway align men ke liye kya kya survey karte hai.

Kis type ke survey hai.

Best alignment kaunse survey me final kiya jata hai.

Isc 3m soil survey kaunse me karte hai.

Kis survey me poora soil profile banaya jata hai.

DPR ke kya kya components hai.

Requirements of ideal roads

Short safe easy & economical.

Surveys for highway alignment

1. Map study
2. Reconaissance
3. Preliminary survey
4. Detailed survey & final location survey

Engg survey

Of finalisation of best alignment is done in preliminary survey.

① Soil survey from 1m to 3m is done in preliminary survey

② In detailed soil survey complete profile of soil is prepared.

Components of DPR. (Detailed proj report)

① Report. ② estimate. ③ Drawings

1.6

- Roadway
  - other name
  - formula.
  - Right of way
    - name
    - formula.

Roadway or formation width.

Roadway = carriageway + shoulder

Road land or right of way or landwidth = Roadway + road margin.

- Road with in m
  - Single lane road
  - two lane road
  - without raised kerb
  - with raised kerb
    - generally kaha detehi
  - multilane pavement
  - Intermediate carriageway
  - width of vehicle.
  - Single deck vehicle ht
  - double decker vehicle ht.

Road width / pavement width (in general)

1. Single lane road : 3.75 m
2. two lane road
   - without raised kerb = 7m
   - with raised kerb = 7.5m (generally used on bridges)
3. Multilane pavement : 3.5 m per lane
4. Intermediate carriageway : 5.5 m

Width of vehicle in general = 2.44 m

Single deck vehicle ht = 3.8 m.

Double deck vehicle ht = 4.7 m

Roadwidths for	plain & rolling terrain	hilly & mountaneous / steep terrain.
1. NH-single & SH lane	12 m	6.25 m
. double lane	12m	8.80 m.
2. MDR . single lane	7m	4.75 m
. double lane	9m	—
3 ODR . single lane	7.5	4.75m
. double lane	9.0	—

(19) Crosssectional element of road

- Caml/cs
- Sh
- Med./div
- ke
- Wi

Crosssectional elements of pavement or road

1) camber / cross slope
2) Shoulder
3) Median / divider.
4) Kerb.
5) Width of pavement

Crossslope/Cambe- purpose.

Camber or cross slope.

- it is provided to drain off rain water/or remove water to prevent entry of surface water into pavement layer.

- Gradient camber relation? G = 2C. gradient = twice of camber

- Cambervalues for heavy rfall & light rfall

	Heavy rainfall	light rainfall
① CC road & high bituminous road	1:50	1:60
② thin bituminous road	1:40	1:50
③ WBM & Gravel road	1:33	1:40
④ Earth road	1:25	1:33

Surface roughness Index
- Bump integ
  - Good pavement value
  - purpose
  - Result kaise deta h?
- International roughness index.
  - kya purpose hy index ka?

Sight distances
- SSD
  - other name
  - what is it?

Surface roughness index

1) Bump integrator

→ good pavement ke liye 150 cm/km hona chahiye value

. The unevenness of pavement surface is commonly measured by using a simple equipment called bump integrator.
. Result it gives vertical undulation of pavement per unit length of road.
Unit → cm/km

2) International roughness index.

. Riding quality of pavement surface is quantified in terms of roughness index IRI.

Sight distances.

SSD: Stopping sight distance or non passing sight distance.
. It is summation of lag or reaction distance + breaking distance

Class 2 4 August 24

SSD

- lag distance kya hai

- Reaction time SSD mekya lete hai

- Brake distance
  - Definition
  - kinetic energy kise equal honi chahiye.

PIEV Theory

lag distance.

lag distance is the distance covered or travelled by vehicle during PIEV reaction time.

time taken reaction time in SSD : 2.5 seconds.

Brake distance : It is the distance travelled by vehicle after application of brake.

- kinetic energy lost should be equal to work done by vehicle.

PIEV theory

1) Perception
2) Intellection
3) Emotion
4) Volition.

SSD

- lag distance ka formula kya h?
  - $1 ms^{-1}$ in km^2
  - lag dist formula in kmph?
- Breaking distance
  - formula in ms^{-1}
  - formula for ascending/descending gradient
  - formula in kmph for gradient

(1) Lag distance (S_l)

S_l : velocity × reaction time

$= [V t_R]$ → sec, → m/s

$1 ms^{-1} = 0.278$ kmph.

$= 0.278 V t_R$ → sec, → kmph

(2) Braking distance (S_b)

work done = kinetic energy loss

$(W.f) * S_b = \frac{V^2}{2g}$

$S_b = \frac{V^2}{2gf}$ ← ms^{-1}

$S_b = \frac{V^2}{2g(f \pm N\%)}$ ← gradient

$S_l = \frac{V^2}{254[f \pm N\%]}$ ← kmph, ← gradient

N is +ve for ascending gradient

N is -ve for descending gradient

f → longitudinal friction coefficient

2.3) SSD

SSD = $S_1 + S_2$

- final formula

$$SSD = 0.278 V t_R + \frac{V^2}{254(f \pm N)}$$

↳ brake effic coeff ki icng

- f ki range & one val.

longitudinal friction

range: 0.35 to 0.40

= 0.35

Note: for 1 lane one way road = SSD

- 1 lane 2 way road ke liy SSD kitna lenge?

for 1 lane two way road

= 2 SSD

- 2 lane 2 way ccd?

for 2 lane 2 way road

: SSD

- ISD?

Intermediate sight distance

= 2 SSD

- HLSD?

Headlight sight distance

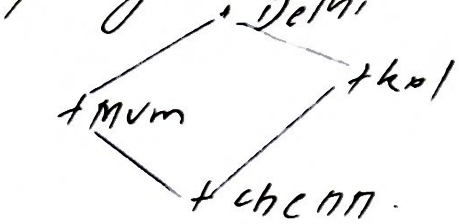
= SSD

NHDP
me kya
kya hai

Imp facet pts

NHDP : National highway developement project/programme

1) Golden Quadrilateral



2) Corridor

(i) N-S corridor [Srinagar - Kanyakumari]

(ii) E-W [Silchar to porbandar]

3) Port connectivity

NRRDA.
- full form.
- PMGSY
  - launch date
  - target

NRRDA : National rural road developement Agency

(i) PMGSY :
Dec 2000

target : population of more than 1000 should be connected to good all weather road. till 2003

2.5] .P7500 ke lie PMGSY me kya tha?

. P > 500 should be connected to above population villags upto 2007.

3rd 20 yr road plan

Third 20 year road plan (Lucknow road plan)

- length of NH formula

length of NH : $\frac{\text{Area in km}^2}{50}$

- length of SH formula

length of SH = $\frac{\text{Area in km}^2}{25}$ or

$62.5 \times$ No of towns

- length of MDR

length of MDR = $\frac{\text{Area in km}^2}{12.5}$

: $30 \times$ No. of towns.

- Expressway target.

Expressway target : 2000 km

NHAI

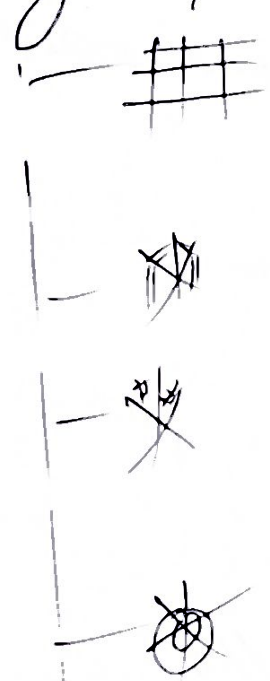
- established yr
- Operatnl yr

NHAI (National highway authority of India). 2.6

. Established 1988

Operational 1995

Road pattern kya kya hai



Road pattern.

1) Rectangular & block pattern
Example Chandigarh.

2) Star & block pattern

3) Star (Radial) & grid pattern.
Eg Nagpur road plan

4) Star (Radial) & circular pattern.
Eg Connaught place Dilli

- Hexagonal kiske liye accha h

5) Hexagonal pattern. : Good for 3 roads meeting @ built up area.

2.7

Concept of Sat
of Syc

- alternate name

- kaise optimum length calculate karte h?

Concept of saturation system

or

maximum utility system

- In this system the optimum road length is calculated on the basis of concept of obtaining maximum utility per unit length of road.

Drawings required

- keymap
  - kya h?
- Index map
  - kya bataya hai

Drawings required

① keymap

The map which shows the proposed & existing roads & imp places to be connected.

② Index map

shows the general topography of the area.

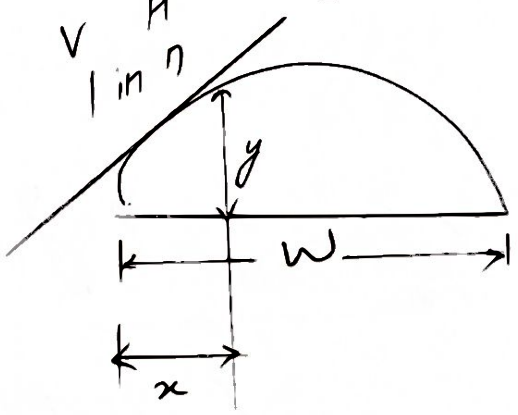
Camber
- purpose
- Shapes
  - Eliptical
    - Good for
    - formula
  - formula

Camber or cross slope

- It prevents entry of surface water or it is provided to drain off rain water

- Shape or types of camber
1) Elliptical or parabolic shape.
Good for bitumin roads or for roads where fast moving vehicles travel

$$y = \frac{2x^2}{nW}$$



V	H
1	n
W/2	nW/2

$$y \propto x^2$$

$$y = \frac{x^2}{n\,W/2} = \frac{2x^2}{nW}$$

- y kya h?
y is rise or height of camber @ x horizontal.

$$\text{Rise} = \frac{2 \times \text{Horizontal}^2}{\text{Slope} \cdot \text{Width}}$$

- W kya h?
W is width of road.

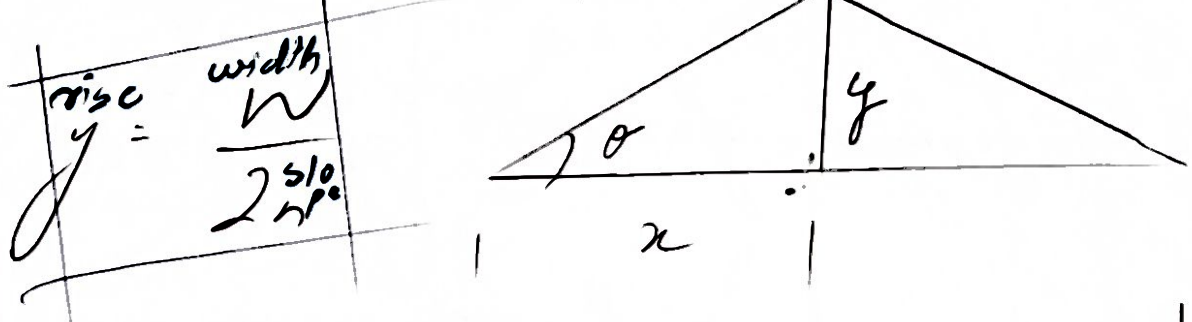
2.7 Camber

- elliptical
  - Disadvantage
  - advantge
- Straight line camber
  - Generally used for:
  - formula

• In this shape vehicles prefer to occupy central portion of roadway hence reduces traffic capacity
• comfortable

linear camber or straight line camber

Generally adopted for c/c road

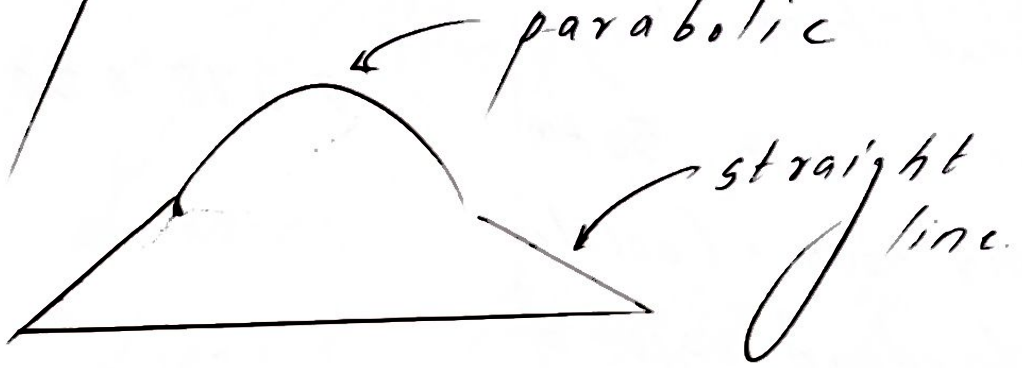


$$\tan\theta = y/x : \frac{1}{n} \Rightarrow y : \frac{x}{n} = \frac{W}{2n}$$

- advantage
  - gives good comfort for slow moving vehicles.
- Disadvantage
  - Discomfort for fast moving vehicle.

Camber shapes
- combinn

Combination of straight line & parabolic camber.



Trafficseperator
- alternate names
- Minimum desirable width.
- Min width bridges.

Traffic separator or median.
- divider / median.
- Minimum desirable width: 5 m.
- for bridges 1.2 m.

Acceleration
- formula for accelern
- formula for retardation

Acceleration or Retardation. / Deceleration.

acceler^n a = fg

deccler^n a = -fg

2. 11

2 Calculate SSD for 1 lane 2 way road if speed of vehicle is 50 km per hour. Consider standard value of coeff of friction drivers reaction time.

Ans: $SSD = 0.278 V t_R + \frac{V^2}{254(f \pm N/100)}$

$= 0.278 \times 50 \times 2.5 + \frac{50^2}{254 \times 0.35}$

$= 62.87$ m.

1 lane 2 way ke liye

$Ans^{SSD}: 2 \times 62.87 = 125.6$.

Tyre Skid

- kyo hota h?
- formula.

Tyre skid

- Due to sudden application of brake skid occurs
- formula $Sl(\%) = \frac{V_v - V_T}{V_v} \times 100\%$.

Tyre slip

- Defn.
- formula

V_v is vehicle velocity (ms^{-1})

V_T is circumferential or tyre velocity

Tyre slip.

- Application of sudden acceleration.
- formula: $S_s(\%) = \frac{V_T - V_v}{V_T} \times 100\%$.

OSD

- alternate names

- di ki value kyah ms⁻¹ me

- t ki. E nah?

- d_3.
  - kyah?
  - formula kyah?

- d_2
  - kiska dist h
  - formula

OSD (passing sight distance/ non stopping sight distance)

overtaking sight distance is summation of three distances

① d_1 : distance travelled by overtaking vehicle (A) while travelling with a speed of overtaken vehicle (B). = $V_b t$: $0.278\ V_b t$ in kmph.
V_b in ms⁻¹

t is reaction time in OSD = 2sec.

② d_3 is distance travelled by opposite coming vehicle in overtaking operation time.

$d_3 = V_c T$ or $0.278\ V_c T$ in kmph.

③ d_2 is distance travelled by overtaking vehicle after accelerating & coming into the front position from B.

d_2 = stats

2.12) OSD

- d_2
  - s formula
    - Default value
  - b
    - kya h?
  - final formula kya h?
    - T ka inverse formula
    - alternate formula for Tsd when S_1 & S_2 are different
  - overtaken ki speed ka formula

s is spacing bt^n vehicles (overtaking & overtaken)

$s = (0.7 V_b + l)$

l is length of wheel base.

$s = 6.1\ m$.

b : distance travelled during acceleration.

$d_2 = 2s + b$

$= \left[\frac{1}{2} a T^2 + V_b T\right]$

$2s = \frac{1}{2} a T^2 \Rightarrow \frac{4s}{a} = T^2 \Rightarrow \boxed{T = \sqrt{\frac{4s}{a}}}$

$T = \sqrt{\frac{2(s_1 + s_2)}{a}}$ $\quad d_2 = s_1 + s_2 + b$

- Jab V_b nai diya h speed of overtaken vehicle tab $V_b = (V_a - 4.5)\ ms^{-1}$

$V_b = (V_a - 16)$ kmph.

OSD
- formula for two way traffic in terms of d
- for one way traffic

Overtaking zone length.
- Min length
- Desirable length
- Minimum OSD ko aur kya bolte hai
- agar minimum overtaking length possible nahi toh kitna deye

Note: for two way traffic

$OSD = d_1 + d_2 + d_3$

for one way traffic

$OSD = d_1 + d_2$

Overtaking zone length.

. Minimum length of overtaking zone: $3 \times OSD$. (matlab jo formula se OSD aya usko 3 times)

. Desirable length of overtaking zone $= 5 \times OSD$

. Minimum OSD is also called safe passing distance.

. if minimum or desirable overtaking zone length is not possible then provide $2 \times SSD$ (overtaking zone length)

3.4 Curves

class 3 6 August 24

Curve
1. Horizontal curve
2. Vertical curve

- Horiz curve
- kahan deteh?

Horizontal curve (superelevation) : is provided in the section of road or pavement where change in direction is required.

- Horizontal curve ka introduction kya karta h.
- radius ko
- centrifugal force ko

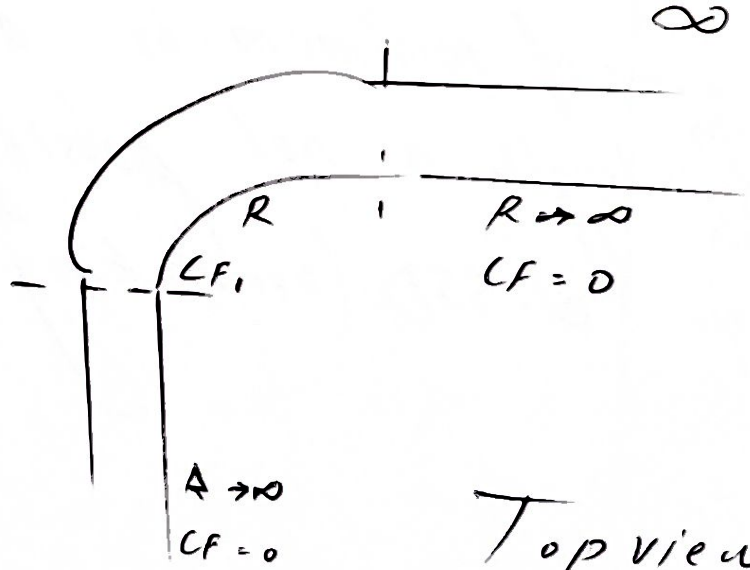
Introduction of horizontal curve gives -

① decreases radius from ∞ to R & then increases to ∞.

② gradual introduction of centrifugal force from 0 to some value then zero

0 - CF - 0
∞ - R - ∞

- Diagram.



Top view.

Horizontal curve & centrifugal force

- kis direction meh?
- kya cause karta h?
- formula

Definition:

1) Centrifugal force on vehicle:

When a vehicle travels in a horizontal curve then an outward force acts on it which creates condition of vehicle to overturn or lateral skid

$$CF = P = \frac{Wv^2}{gR} = \frac{Wt \times vel^2}{gacc \times Rad}$$

W = wt of vehicle in kg

v is velocity of vehicle in ms^{-1}

R is radius of circular curve in metre.

g is gravitational acceleration in ms^{-2}.

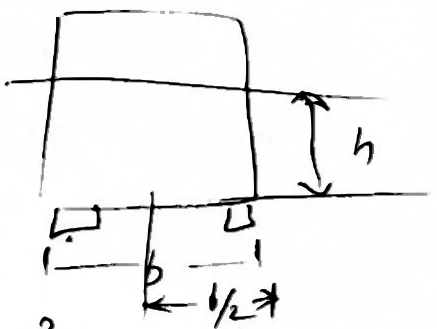
- Impact factor
  - alternate name
  - formula

(alternate name)

2) Impact factor or centrifugal ratio:

$$\frac{P}{W} = \frac{v^2}{gR}$$

- Conditions
  - for ⇌
    - formula
    - b kyah
    - h kyah



Condition

1) for ⇌ $\frac{P}{W} = \frac{v^2}{gR} = \frac{b}{2h}$

b is width of vehicle &

h is height of CG of vehicle or ht of centrifugal force acting from road surface

3.3 Conditions in horiz curve

(2) to avoid over turning

(3) to avoid lateral skid

(ii) to avoid overturning

$$\frac{b}{2h} > \frac{P}{W}$$

(iii) to avoid lateral skid

$$f_{lateral} > P/W$$

for the condition of sliding no (skidding) and no overturning

$$f_{lateral} > P/W \quad \& \quad b/2h > P/W$$

- for no sliding & no overturning (formula)

- flat pavement me curve pe vehicle pe kahan pressure zyada ayega

- inner wheels pe pressure zero kab hoga.

Note: ① if the pavement is kept flat then the pressure on outer wheels of vehicle will be more than pressure on inner wheels due to centrifugal force.

② for limiting condition if overturning occurs then pressure ② inner wheel becomes equal to zero. (generally for flat roads)

Horiz curve is superelevn

— kya h?

— Kyu h?

— Kaise h?

Superelevation - e or cant

- Rise of outer edge of pavement wrt inner edge on horizontal curve it is called superelevation or cant or banking.
- It is a transverse slope.
- $e = \tan\theta = \frac{x}{y}$

① General equation with fric & superelvn

1) General equation (with friction & superelevation)

$$\left(e + f_{lateral}\right) = \frac{v^2}{gR}$$

② equilibrium superelevn formula

2) Equilibrium superelevation ($f_{lateral} = 0$)

$$e = \frac{v^2}{gR} = e_{eq} = \frac{V^2}{127R}$$

— flat kitna h?

for this condition coefficient of Lateral friction is considered zero.

③ allowable speed without superelevn

— v formula

3) allowable speed of vehicle on a horiz curve without superelevation. ($e = 0$)

$$f_{lateral} = \frac{v^2}{gR} \Rightarrow \boxed{v = \sqrt{f_{lat} \cdot g \cdot R}}$$

$v_{allowable}$ = speed

Superelevn design steps

then superelevation should be provided e. (chote wale ko denge) (3.6

emax ki value IRC se milta hai.

(ii) if e > emax (chote wale ko denge) then provide superelevation emax.

Step ③ →

Calculate friction from general eqn

$$f_{flat} = \frac{V^2}{127R} = \frac{v^2}{gR}$$

$$f_{flat} = \frac{V^2}{127R} - e$$

(ii) if flat < 0.15 (codal value) ok (chote wale ko denge)

put road friction as flat calculated in step 3

(ii) if flat > 0.15 Not ok (chote wale ko denge)

if not ok the speed should be restricted & provide flat = 0.15

- If e > emax toh kya karenge
- Step ③ - f calc.
- flat < 0.15
- flat > 0.15
- Step ④